

Module Lattice Based Digital Signature Standard (ML-DSA)

ML-DSA is a lattice-based digital signature algorithm standardized by NIST's FIPS 204 publication that provides a quantum resistant method to generate and verify digital signatures. It's set to replace current digital signature algorithms such as RSA (Rivest-Shamir-Adleman) and ECDSA (Elliptic Curve Digital Signature Algorithm) that are quantum vulnerable.

ML-DSA is based on the hardness of the MLWE (Module Learning With Errors) problem, which is itself based on the idea of inferring a linear function over noisy data. The problem is considered to be quantum resistant, which carries over to ML-DSA.